# HVDC System Design

ARPA-E

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# System Studies

HVDC design process and simulation packages

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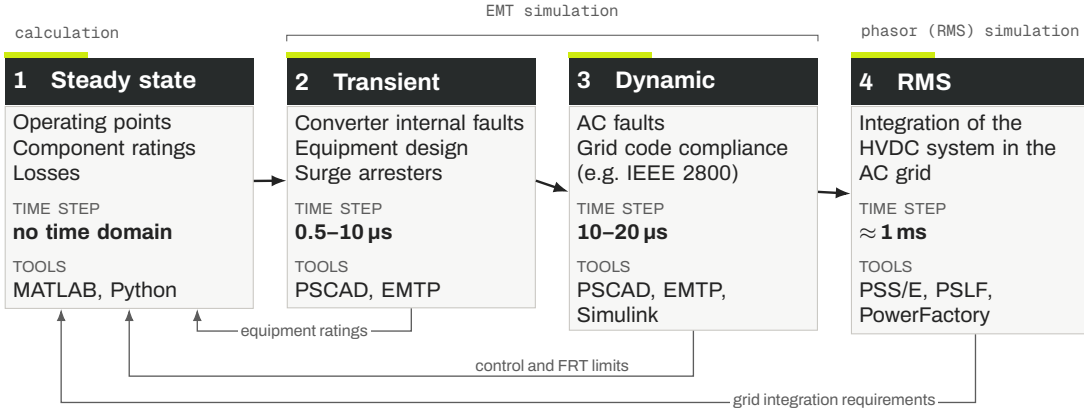
# Abbreviations

| | |
|---|---|
| AC | Alternating current |
| DC | Direct current |
| EMT | Electromagnetic transient |
| EMTP | Electromagnetic Transients Program |
| FRT | Fault ride-through |
| HVDC | High-voltage direct current |
| IEEE | Institute of Electrical and Electronics Engineers |
| PSCAD | Power Systems Computer Aided Design |
| PSLF | Positive Sequence Load Flow |
| PSS/E | Power System Simulator for Engineering |
| RMS | Root mean square |

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# Design process
# System studies

Results of each study feed back into the steady-state design.




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# Steady state
# Operating points, ratings and losses

## Calculation, no simulation tool

Algebraic calculation of the converter station at its steady-state operating points.

- Operating points across the power, AC voltage and DC voltage range
- Steady-state rating of the components
- Losses per component and for the station
- Starting point for all following studies

| | |
|---|---|
| **Tools** | MATLAB, Python |
| **Time step** | no time domain |
| **Results** | Operating points, component ratings, losses |

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# Transient simulation
# Converter internal faults

## EMT simulation, detailed model

Transient behaviour of the system during converter internal faults.

- Converter internal faults, e.g. valve and converter bus faults
- Design of the equipment for the transient stresses
- Design of protective equipment such as surge arresters
- Returns updated ratings to the steady-state design

| | |
|---|---|
| **Tools** | PSCAD, EMTP |
| **Time step** | 0.5–10 µs |
| **Results** | Equipment stresses, arrester ratings |

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# Dynamic simulation
# AC faults and grid code compliance

## EMT simulation, control-oriented model

Dynamic behaviour of the system during AC faults.

- AC faults at and near the point of connection
- Grid code compliance defined by e.g. IEEE 2800
- FRT performance and control response
- Returns control and FRT limits to the design

| | |
|---|---|
| **Tools** | PSCAD, EMTP, Simulink |
| **Time step** | 10–20 µs |
| **Results** | Control performance, FRT behaviour, compliance evidence |


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# RMS model
# Integration in the AC grid

## Phasor simulation, grid-level model

Simulation of the integration of the HVDC system in the AC grid.

- HVDC system as part of the wider AC network
- Model handed over to the grid operator for planning studies
- Returns grid integration requirements to the design

| | |
|---|---|
| **Tools** | PSS/E, PSLF, PowerFactory |
| **Time step** | ≈ 1 ms |
| **Results** | RMS model for grid studies |

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